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NLP Postdoctoral Research Fellow

Natural Language Processing | Text Simplification | Readability | Explainable AI

Profile

PhD-qualified NLP researcher with over 15 years of experience in computational linguistics, text simplification, readability analysis and applied language technology. My central question is how complex public-facing information can be made easier to understand and use without losing essential meaning. Across the Horizon Europe iDEM project, UCREL and my doctoral research, I develop multilingual corpora, annotation frameworks, simplification pipelines, web-based research software and APIs, and explainable, uncertainty-aware models for accessible and reliable communication.

Research Expertise & Technical Skills

Research focus

Easy-to-Read text; readability and sentence complexity; lexical, syntactic and semantic simplification; accessible communication; multilingual and under-resourced NLP.

Modelling and evaluation

Transformers and LLMs; controlled generation; classification and regression; explainable AI; uncertainty estimation; denoising; error analysis.

Corpus and resource development

Corpus construction, cleaning and alignment; annotation schemes; linguistic feature engineering; Sketch Engine resources; dataset documentation and release.

Programming and data

Python; R; C#; SQL; PyTorch; Hugging Face; scikit-learn; pandas; spaCy; NLTK; Gensim; Oracle databases.

Web and research software

API and web-based NLP tool development; Flask applications; interactive analysis and visualisation tools; .NET web application development.

Infrastructure and reproducibility

Docker; Git/GitHub; Linux and command-line workflows; reproducible experimental pipelines; open-source release and technical documentation.

Selected Research Contributions, Software & Open Outputs

Accessible multilingual simplification iDEM / 2024–present
Lead research on readability, complexity typology and LLM simplification/evaluation. Contributed to multilingual Easy-to-Read resources for democratic participation and meaning-preserving adaptation.
 Democracy Made Easy  E2R corpus paper

Reliable readability modelling and explanation 2025–2026
Developed explainability, denoising and uncertainty-estimation analyses that identify why texts are difficult and when a multilingual classifier should abstain from prediction. Released reproducible resources and experimental code.
 why-tough: readability explanation dataset  denoising-difficulty  To Predict or Not to Predict

Arabic simplification and readability PhD / 2021–2023
Developed a corpus-based automatic Modern Standard Arabic text simplification system and released Arabic readability and CEFR-level lexical resources supporting classification, vocabulary profiling and simplification research.
 PhD thesis  Arabic Readability Corpus  Arabic CEFR List

FreeTxt multilingual survey-analysis software UCREL / 2022–2025
Led FreeTxt software prototype development, turning bilingual corpus-analysis methods into a web-based toolkit for qualitative survey analysis, visualisation and cultural-heritage research.
 FreeTxt app  FreeTxt-Flask

Welsh thesaurus and low-resource language resources UCREL / 2022–2024
Served as sole software prototype developer for Thesawrws and co-developed methods for an interactive open-source thesaurus for contemporary Welsh.
 Welsh Thesawrws

Open datasets, code and reproducible NLP pipelines Ongoing
Released annotation frameworks, corpora, experimental code and technical documentation for text simplification, sentence difficulty, uncertainty estimation, denoising and readability evaluation.
 XAI dataset  Arabic corpus  Denoising experiments

Research Experience

06/2024 – present	NLP Postdoctoral Research Fellow <i>Horizon Europe – iDEM Project: Innovative and Inclusive Democratic Spaces</i> Lead NLP research on sentence-level readability, complexity classification and text simplification for accessible democratic and institutional information. Develop LLM-based simplification and evaluation pipelines, complexity-typology analysis, interpretability, data denoising and uncertainty-aware assessment; contribute to multilingual original/Easy-to-Read corpora and reusable APIs/resources.	University of Leeds, CTIS
10/2022 – 05/2024	NLP Research Associate <i>FreeTxt, Thesawrws, Welsh Digital Grid and Environmental Change Analysis projects</i> Led software prototype development for FreeTxt, turning bilingual corpus-analysis methods into a web-based toolkit for academic and GLAM-sector users. Served as sole software prototype developer for Thesawrws and co-developed methods for an interactive open-source thesaurus for contemporary Welsh.	Lancaster University, UCREL Group

07/2022 – 2024	Research Assistant <i>Arabic Teaching Council</i> Supported Arabic teaching in Leeds and West Yorkshire through outreach, workshops, professional development events and mentoring coordination.	University of Leeds
03/2022 – 07/2022	Research Assistant – Corpus Builder <i>Arabic-English constitutions corpus project</i> Built, aligned and semi-automatically verified Arabic-English constitutional legal corpora for publication on Sketch Engine.	University of Leeds
05/2020 – 07/2021	Research Assistant – Corpus Annotator <i>Arabic Semantic Annotator based on the UCREL USAS Semantic Tagger</i> Researched and annotated a large Arabic semantic lexicon and produced analysis outputs for project presentations.	Lancaster University, Zayed University and NYU Abu Dhabi
01/2021 – 03/2021	Research Assistant – Data Analyst <i>Arabic learners' perceptions of grammar knowledge</i> Conducted qualitative and quantitative analysis of approximately 264 questionnaire responses.	University of Leeds
10/2020 – 01/2021	Research Assistant – Tweets Annotator <i>Influenza and COVID-19 Arabic Twitter corpus</i> Analysed and annotated approximately 2,000 Arabic tweets for public-health discourse categories.	Lancaster University
12/2010 – 06/2014	Developer and Corpus Annotator <i>ALTEC diacritised and named-entity annotated corpora</i> Developed annotation workflows and interfaces and supported corpus annotation across 104 named-entity categories.	Alexandria University / RDI

Teaching & Supervision Experience

2024 – 2026	Teaching Support <i>MODL5089M Machine Translation and Natural Language Processing (40493)</i> Contributed to teaching Natural Language Processing for translation, supporting student learning in machine translation and NLP concepts and methods.	University of Leeds
2025 – present	Undergraduate Research Supervisor <i>Explainable AI, readability evaluation and text simplification</i> Secured and led two consecutive N8 CIR Summer Internship proposals, supervising undergraduate research on accessibility-oriented NLP.	N8 CIR / University of Leeds
09/2020 – 2022	Teaching Assistant <i>Essential Skills in Practical Arabic</i> Supported language teaching, practical sessions, assessment and student learning in Arabic language provision.	University of Leeds
06/2011 – 09/2018	Demonstrator / Associate Lecturer <i>Phonetics and Linguistics / Language Applications Departments</i> Taught Natural Language Processing, Language Technology, Corpus Linguistics, Applied Linguistics, Specialised Dictionaries and Programming.	Alexandria University
2017 – 2020	Teaching and Student Support Roles <i>Research skills; Arabic as a Foreign Language; academic writing support</i> Delivered research-skills support for postgraduate students and supported Modern Standard Arabic and Egyptian Colloquial Arabic teaching for international learners.	Alexandria University / University of Leeds

Education

2019 – 2023	PhD in Natural Language Processing <i>Newton-Mosharafa Fund / British Council PhD Research Scholarship</i> – Thesis: <i>An Automatic Modern Standard Arabic Text Simplification System: A Corpus-Based Approach.</i>	University of Leeds, United Kingdom
2011 – 2016	MSc in Computational Linguistics <i>GPA: 3.8</i> – Thesis: <i>Towards Arabic Named Entity Recognition.</i>	Alexandria University, Egypt
2011 – 2012	Professional Scholarship in Web Application Development <i>Grade: Excellent; MCPD and MCTS certifications</i>	RAYA Academy for IT / MCIT, Egypt
2006 – 2010	BA in Phonetics and Linguistics <i>Very Good with Honours</i> – Graduation project: intelligent online website for teaching Modern Standard Arabic as a second language.	Alexandria University, Egypt

Awards & Professional Development

Enrichment Student Community Award Alan Turing Institute, Cohort 2022–2023	Lancaster Summer School in Corpus Linguistics Lancaster University, 2020
SICSS-Edinburgh Summer School University of Edinburgh, 2021	Lisbon Machine Learning School (LxMLS) Instituto Superior Técnico, 2020
Research Software Engineering & Digital Humanities Summer School Alan Turing Institute, 2021	Computational Humanities Summer School & OpenMed Training Heidelberg University / Erasmus+, 2017–2019

Academic Leadership, Research Community & Knowledge Exchange

- **Principal Investigator / Internship Supervisor**, N8 CIR Digital Research Internship Programme: *What Do Large Language Models Know? Text Simplification Analysis* (2025) and selected 2026 project *Interpreting Readability Evaluation with Explainable AI*; supervised research at the interface of accessibility, explainable AI and reproducible NLP.
- **Co-editor**, *Proceedings of the Fourth Workshop on Text Simplification, Accessibility and Readability (TSAR 2025)*, Association for Computational Linguistics.
- **Co-editor**, *Proceedings of the First International Conference on Natural Language Processing and Artificial Intelligence for Cyber Security (NLPACS 2024)*, ACL Anthology.
- **Peer reviewer / programme committee member** for NLP venues; contributor to open science through datasets, GitHub code, annotation frameworks, simplification software and reproducible experiment releases.
- **Teaching/Organising Team**, UCREL Summer School in Corpus Linguistics, Lancaster University (2024): practical sessions on text visualisation and summarisation, Docker for NLP 101, and a mini hackathon/team collaboration challenge.
- **Conference organisation**: Organising Committee, Arabic Linguistics Forum (University of Leeds, 2020); Lead Organiser, LCS Postgraduate Conference (University of Leeds, 2019); Founder/Organiser, Postgraduate Students' Scientific Conference (Alexandria University, 2012 and 2017).

Publications

2026

- **Khallaf, N.**, & Sharoff, S. (2026). *How Much Noise Can BERT Handle? Insights from Multilingual Sentence Difficulty Detection*. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 6132–6143. ELRA. Proceedings.
- **Khallaf, N.**, & Sharoff, S. (2026). *To Predict or Not to Predict? Towards Reliable Uncertainty Estimation in the Presence of Noise*. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 152–168. ELRA. Proceedings.
- Riegler, V., Bott, S., Saggion, H., Rascón Alcaina, A., & **Khallaf, N.** (2026). *A Multilingual Human Annotated Corpus of Original and Easy-to-Read Texts to Support Access to Democratic Participatory Processes*. In *Proceedings of LREC 2026*, 1117–1128. ELRA. Proceedings.
- Hilaraca, K., **Khallaf, N.**, & Sharoff, S. (2026). *Align and Shine: Building High-Quality Sentence-Aligned Corpora for Multilingual Text Simplification*. In *Proceedings of the 19th Workshop on Building and Using Comparable Corpora (BUCC) @ LREC 2026*, 62–71. ELRA. Proceedings.
- Astier, C., **Khallaf, N.**, Barriuso, O., Mazzanti, C., Sayman, V., Bott, S., Sharoff, S., & Saggion, H. (2026). *Lost in Deliberation: Making Democracy Understandable*. In I. Koprinska, J. Mendes-Moreira, & P. Branco (Eds.), *Machine Learning and Principles and Practice of Knowledge Discovery in Databases*. ECML PKDD 2025. Communications in Computer and Information Science, vol. 2843, 427–444. Springer. DOI.

2025

- Hayakawa, A., **Khallaf, N.**, Saggion, H., & Sharoff, S. (2025). *UoL-UPF at TSAR 2025 Shared Task: A Generate-and-Select Approach for Readability-Controlled Text Simplification*. In *Proceedings of TSAR 2025*, 193–210. ACL. ACL Anthology.
- **Khallaf, N.**, Bott, S., Eugeni, C., O'Flaherty, J., Sharoff, S., & Saggion, H. (2025). *Democracy Made Easy: Simplifying Complex Topics to Enable Democratic Participation*. In *Proceedings of AI & EL/PL*, 108–124. EAMT. ACL Anthology.
- **Khallaf, N.**, Ezeani, I., Knight, D., Rayson, P., El-Haj, M., Vidler, J., Davies, J., & Alva-Manchego, F. (2025). *FreeTxt: Analyse and Visualise Multilingual Qualitative Survey Data for Cultural Heritage Sites*. In *Proceedings of RANLP 2025*, 541–545. INCOMA Ltd. ACL Anthology.
- **Khallaf, N.**, Eugeni, C., & Sharoff, S. (2025). *Reading Between the Lines: A Dataset and a Study on Why Some Texts Are Tougher Than Others*. In *Proceedings of WRAICOGS 2025*, 24–34. ICCL. ACL Anthology.

2024–2023

- Knight, D., **Khallaf, N.**, Rayson, P., El-Haj, M., Ezeani, I., & Morris, S. (2024). *FreeTxt: A Corpus-Based Bilingual Free-Text Survey and Questionnaire Data Analysis Toolkit*. *Applied Corpus Linguistics*, 4(3), 100103.
- El-Farahaty, H., **Khallaf, N.**, & Alonayzan, A. (2024). *بناء المدونات اللغوية القانونية الخاصة بجامعة ليدز: منهجيات وتحديات* [Building the University of Leeds Legal Corpora: Methodologies and Challenges]. *Majallat al-Lisaniyat al-Arabiyya*, 18, 96–116.
- **Khallaf, N.**, Arfon, E., El-Haj, M., Morris, J., Knight, D., Rayson, P., Hammouda, T. H., & Jarrar, M. (2023). *Open-Source Thesaurus Development for Under-Resourced Languages: A Welsh Case Study*. In *Proceedings of LDK 2023*, 306–315.
- El-Farahaty, H., **Khallaf, N.**, & Alonayzan, A. (2023). *Building the Leeds Monolingual and Parallel Legal Corpora of Arabic and English Countries' Constitutions: Methods, Challenges and Solutions*. *Corpus Pragmatics*, 7, 103–119.
- **Khallaf, N.** (2023). *An Automatic Modern Standard Arabic Text Simplification System: A Corpus-Based Approach*. Doctoral thesis, University of Leeds. White Rose eTheses.

Selected Earlier Publications

- **Khallaf, N.**, Sharoff, S., & Soliman, R. (2022). *Towards Arabic Sentence Simplification via Classification and Generative Approaches*. In *Proceedings of WANLP 2022*, 43–52. ACL Anthology.
- El-Haj, M., de Souza, E., **Khallaf, N.**, Rayson, P., & Habash, N. (2022). *AraSAS: The Open-Source Arabic Semantic Tagger*. In *Proceedings of OSACT 2022*.
- **Khallaf, N.**, de Souza, E., El-Haj, M., & Rayson, P. (2022). *Semantic Domains across Topics, Genders, and Languages*. In D. Palfreyman & N. Habash (Eds.), *Bilingual Writers and Corpus Analysis* (pp. 96–120). Routledge.
- **Khallaf, N.**, & Sharoff, S. (2021). *Automatic Difficulty Classification of Arabic Sentences*. In *Proceedings of WANLP 2021*, 105–114. ACL Anthology.
- **Khallaf, N.**, & Alansary, S. (2013). *Towards Arabic Named Entity Recognition System*. In *The Thirteenth Conference on Language Engineering (ESOLEC)*, 217–229. Cairo, Egypt.